

EXPERIMENT 18

DIGITAL SIGNATURE

AIM

To implement **RSA digital signature generation and verification** in C and verify the authenticity and integrity of a message.

ALGORITHM

Signature Generation

1. Start the program.
2. Generate an RSA private key and corresponding public key.
3. Read the message.
4. Generate the SHA-256 hash of the message.
5. Sign the hash using the RSA private key.
6. Display the generated digital signature.

Signature Verification

1. Use the RSA public key to verify the signature.
2. Compare the calculated message hash with the signed hash.
3. If they match, the signature is valid.
4. Otherwise, the signature is invalid.
5. Stop the program.

CODE

```
#include <stdio.h>
```

```
#include <string.h>
```

```
long long power_mod(long long base, long long exp, long long mod)
```

```
{
```

```
    long long result = 1;
```

```
    while (exp > 0)
```

```

    {
        if (exp % 2 == 1)

            result = (result * base) % mod;

        base = (base * base) % mod;

        exp = exp / 2;
    }

    return result;
}

/* Simple message hash for demonstration */

long long hash_message(char message[])
{
    long long hash = 0;

    for (int i = 0; message[i] != '\0'; i++)
    {
        hash = (hash + message[i]) % 3233;
    }

    return hash;
}

Int main()
{
    char message[1000];

    /* RSA parameters */

```

```
long long n = 3233;

long long e = 17;

long long d = 2753;

long long hash;

long long signature;

long long verified_hash;

printf("Enter the message: ");

fgets(message, sizeof(message), stdin);

message[strcspn(message, "\n")] = '\0';

/* Generate message hash */

hash = hash_message(message);

printf("\nMessage Hash: %lld", hash);

/* Generate digital signature */

signature = power_mod(hash, d, n);

printf("\nDigital Signature: %lld", signature);

/* Verify signature */

verified_hash = power_mod(signature, e, n);

printf("\nVerified Hash: %lld", verified_hash);

if (hash == verified_hash)

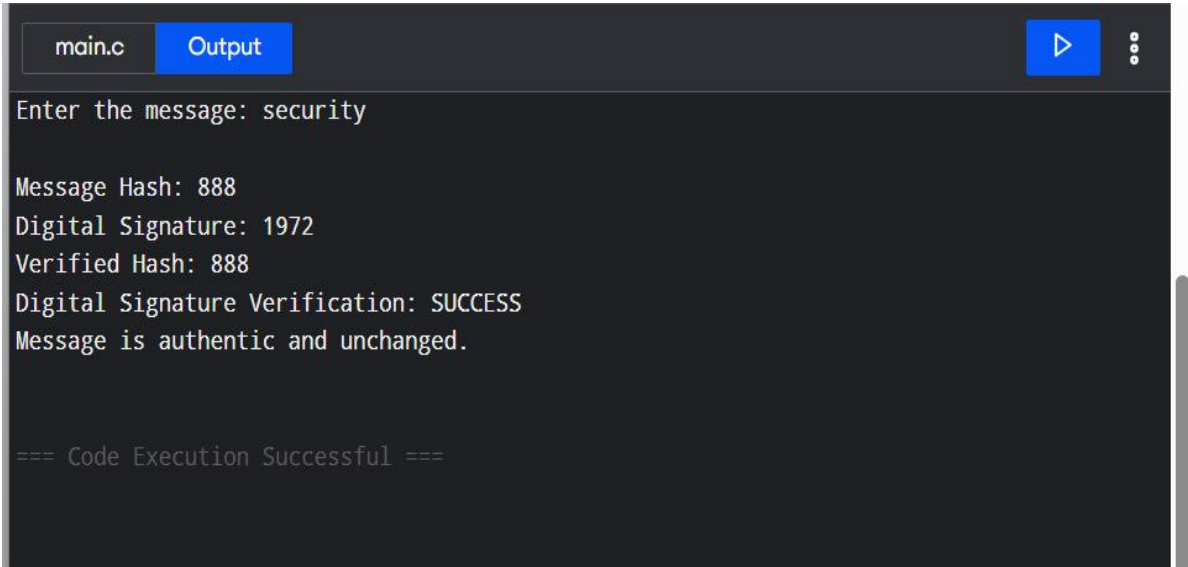
{

    printf("\nDigital Signature Verification: SUCCESS");

    printf("\nMessage is authentic and unchanged.\n");
```

```
}  
  
else  
  
{  
  
    printf("\nDigital Signature Verification: FAILED");  
  
    printf("\nMessage may have been modified.\n");  
  
}  
  
return 0;  
  
}
```

OUTPUT



```
main.c  Output  ▶  ⋮  
Enter the message: security  
  
Message Hash: 888  
Digital Signature: 1972  
Verified Hash: 888  
Digital Signature Verification: SUCCESS  
Message is authentic and unchanged.  
  
=== Code Execution Successful ===
```

RESULT

Thus, the **RSA digital signature** was successfully generated and verified in C. The successful verification confirms the **authenticity and integrity** of the message.